
Measurable Writing Standards for AI-Native Venues

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Abstract

1 Language models now draft manuscripts, assist review, and screen submissions, so
2 the institutions of publishing are becoming AI-native. We argue that this transition
3 inherits a communication problem that venues have not yet measured, and one
4 that AI tooling does not correct. We analyze roughly 2.8 million arXiv papers
5 (1991 to 2025), 30,595 NeurIPS papers (1987 to 2025), and 24.5 million PubMed
6 abstracts (1990 to 2025) using classical readability formulas, nine writing style
7 metrics, acronym density and reuse, citation counts, and an LLM as judge protocol.
8 We report four findings. NeurIPS abstracts have become harder to read on every
9 classical metric, with Flesch Reading Ease falling from approximately 24 in 1987
10 to approximately 8 in 2025 and sensational language rising by approximately 80
11 percent between 2015 and 2025. Acronym density rose from 0.33 to 4.15 per
12 100 words in titles and from 0.62 to 2.97 in abstracts, diverging from the non-ML
13 baseline after 2016, and roughly 89 percent of ML acronyms are used fewer than
14 ten times. More readable NeurIPS papers tend to be more cited, an association that
15 survives a control for subfield. Crucially for an AI-native venue, the AI tools meant
16 to help do not correct these trends. Six language models used as readability judges
17 report that readability is flat or improving, in direct contradiction to every classical
18 metric, and the divergence widens after 2022. We propose seven measurable,
19 auditable standards that AI-native venues could pilot, and we argue that automated
20 screening must be anchored in transparent human-grounded metrics rather than a
21 language model’s own judgment.

1 Introduction

23 Academic publishing is becoming AI-native, and the shift is far enough along to be measured rather
24 than anticipated. Model assistance in scientific writing had reached up to 22 percent of computer
25 science arXiv abstracts by September 2024 [Liang et al., 2025b], with comparable shares documented
26 across institutional and consumer text [Liang et al., 2025a]. On the open web, a detector based
27 audit of 43,000 Common Crawl URLs places the point at which machine generated articles overtook
28 human written ones in late 2024 [Druck et al., 2025]. Assessment is moving the same way, and a
29 language model scoring written text is now a standard evaluation instrument [Zheng et al., 2023,
30 Liu et al., 2023, Gu et al., 2026]. Venues have begun to deploy that tooling directly. Every main
31 track submission to AAAI 2026 received one identified machine generated review, 22,977 papers
32 processed in under a day [Biswas et al., 2026]. The governance of authorship, review and screening
33 is therefore being rebuilt now, and the measurement standards it encodes are being chosen now.

34 We argue that this transition inherits a communication problem that venues have not yet had to
35 account for, and that AI tooling does not correct. That tooling enters the pipeline in two distinct roles.
36 In the first, models help write the text. In the second, models are asked to judge it. This paper is
37 about the second, where a model decides what counts as good writing.

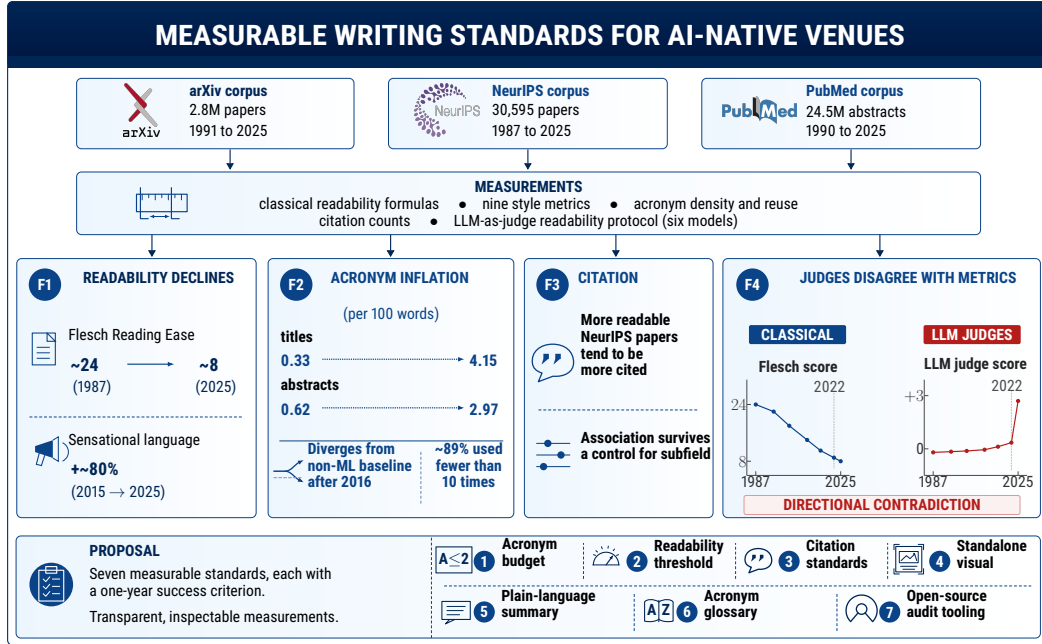


Figure 1: **Overview.** Three corpora are measured with one set of metrics, yielding four findings (F1 to F4) and seven proposed standards. Classical metrics and LLM judges disagree in direction over the same abstracts (F4), which is the result the standards are designed around.

Across the four decades this paper measures, acronym density in machine learning rose, abstracts became harder to read on every classical measure, and submissions grew close to exponentially [Chen et al., 2025]. Jeschke et al. [2019] describe the general version of this pattern as knowledge in the dark, where findings exist in the literature but do not reach the people who could use them. Medical publishing has debated the same ground for decades, and abbreviation use there is governed by explicit editorial guidance of a kind machine learning has no equivalent for [Berlin, 2013]. Machine learning has not adopted comparable venue-level norms, and the measurement standards its automated pipeline will encode remain undetermined. Submission growth is what makes automated screening attractive to venues [Chen et al., 2025, Biswas et al., 2026], and first-pass assessment of writing quality is among the tasks such a system would be given.

We test the communication problem empirically across roughly 2.8 million arXiv papers (1991 to 2025), 30,595 NeurIPS papers (1987 to 2025), and 24.5 million PubMed abstracts (1990 to 2025). arXiv papers are classified as machine learning if their primary category is one of `cs.LG`, `cs.AI`, `cs.CV`, `cs.CL`, `cs.NE`, `cs.RO`, `cs.IR`, `cs.MA`, `cs.HC`, `cs.CY`, or `stat.ML`. All remaining categories form a non-ML control group, so machine learning trends can be separated from field-wide trends without confounding venue and domain.

We apply five families of metrics: 15 classical readability scores, the nine writing style metrics of Hohmann et al. [2025], the acronym extraction algorithm of Barnett and Doubleday [2020], citation counts for NeurIPS papers from OpenAlex and Semantic Scholar, and the LLM as judge protocol of Cachola et al. [2025]. Of these, readability, style, and judge scores are computed on abstracts only, while acronym density is reported separately for titles and for abstracts. Each measurable stands in for a cost the reader pays, and Appendix B gives the cognitive evidence for each.

We make three contributions.

- We assemble measurable evidence that machine learning communication norms have degraded relative to a controlled baseline (Sections 2 to 4).
- We then show that the AI tools a venue would reach for do not detect that degradation, and report it in the opposite direction (Section 5).
- We finally translate the evidence into seven measurable, auditable standards (Section 6).

66 The publishing pipeline runs from author to submission to reviewer to program committee to platform.
 67 Our result concerns measurement validity at the reviewer and committee nodes, and the standards act
 68 at the author and platform nodes.

69 2 Reading Difficulty

70 **Classical readability.** The Flesch Reading Ease score assigns values from 0 to 100, where higher
 71 values indicate easier reading [Flesch, 1948, DuBay, 2004]. On the bands Flesch [1979] gives, a
 72 score between 30 and 10 is very difficult and best understood by university graduates, and a score at
 73 or below 10 is extremely difficult and falls in the band he labels professional. Plavén-Sigra et al.
 74 [2017] report the measure declining across the sciences. We use it as the primary indicator because
 75 its scale carries a direct empirical reading in terms of the education level a text demands. We note
 76 that the 15 are not independent instruments. They share surface inputs, and over NeurIPS their mean
 77 absolute pairwise correlation is 0.68 with the first principal component accounting for 72 percent of
 78 the variance. We therefore read their agreement as one well attested direction of change rather than
 79 as 15 confirmations. The full 15-metric panels appear in Appendix C.

80 Figure 2 shows Flesch Reading Ease on NeurIPS, arXiv ML, arXiv non-ML, and PubMed from 1987
 81 to 2025. NeurIPS falls from approximately 24 in 1987 to approximately 8 in 2025, a sustained decline
 82 across 38 years that carries the corpus from the university graduate band into the professional one,
 83 and the arXiv ML aggregate follows a similar path. The arXiv non-ML baseline declines more slowly,
 84 which confirms that the accelerated decline is specific to machine learning rather than a universal
 85 property of scientific writing. All 15 classical metrics move in the same direction on machine learning
 86 venues, and the rate of decline steepens after the late 2022 availability of instruction-tuned writing
 87 assistants, marked in the figure. P2 in Section 6 sets a readability floor against this decline.

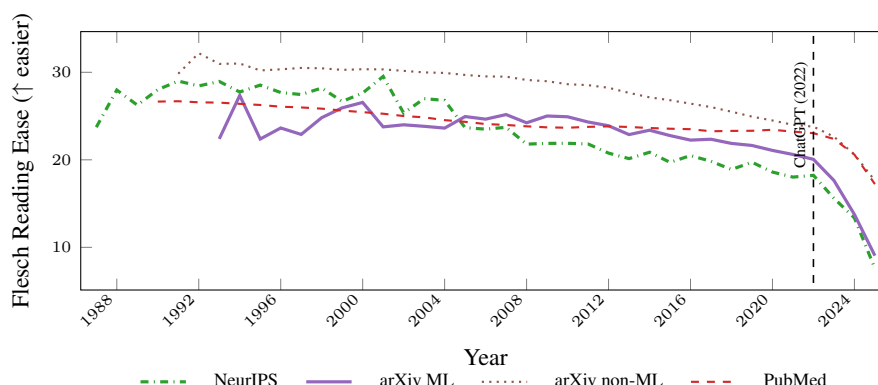


Figure 2: **Flesch Reading Ease on NeurIPS, arXiv ML, arXiv non-ML, and PubMed, 1987–2025.** Higher values indicate easier reading (↑). NeurIPS falls from approximately 24 in 1987 to approximately 8 in 2025. arXiv ML follows a similar trajectory. The arXiv non-ML baseline declines more slowly, indicating that the accelerated decline is specific to ML venues. Vertical dashed line marks the late 2022 availability of widely-used instruction-tuned writing assistants.

88 **Sensational language.** Sensational language compounds the picture. We count the eight hype
 89 categories of Millar et al. [2022] and a ninth added by Hohmann et al. [2025]. Figure 3 shows total
 90 sensational language per 100 abstract words. The NeurIPS rate rises by approximately 80 percent
 91 between 2015 and 2025, from 1.10 to 1.98 per 100 words, and the divergence between machine
 92 learning and non-ML venues accelerates after 2022. Across the broader set of style metrics the
 93 machine learning and non-ML families are largely parallel rather than divergent (Figure 8), so the
 94 bulk of the readability decline appears in the classical readability and sensational language families.
 95 The nine categories are shown separately in Figure 9. Hohmann et al. [2025] observe abrupt shifts in
 96 individual journals’ writing style, which they suggest may reflect changes in editorial practice.

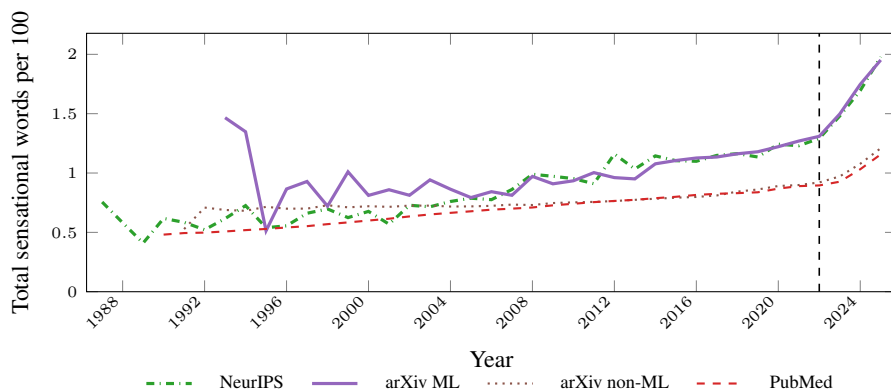


Figure 3: **Total sensational language per 100 abstract words, 1987–2025.** The NeurIPS total rate rises by approximately 80 percent between 2015 and 2025 (from 1.10 to 1.98 per 100 words). The arXiv non-ML baseline rises far more slowly. The divergence between ML and non-ML venues accelerates after the late 2022 availability of instruction-tuned writing assistants (vertical dashed line).

3 Acronyms

Density. Jargon disrupts fluent processing of scientific text even when definitions are supplied [Shulman et al., 2020], and an acronym the reader has not met before is jargon by construction. Figure 4 reports acronym density for titles and abstracts. NeurIPS title density rose from 0.33 per 100 words in 1987 to 4.15 in 2025, more than a twelvefold increase, and abstract density rose from 0.62 to 2.97 over the same period. Density per 100 words is hard to picture for a title, so we also report counts. The mean NeurIPS title carried 0.03 acronyms in 1987 and 0.43 in 2025, and the share carrying at least one rose from 3 to 35 percent. arXiv ML rose from 1.60 to 4.64 in titles and from 1.33 to 3.13 in abstracts between 2007 and 2025. The arXiv non-ML control rose far less, from 1.40 to 2.13 in abstracts, and stays between roughly 2.1 and 2.8 in titles, bracketing the science-wide figure of 2.40 that Barnett and Doubleday [2020] report for 2019. Machine learning and the non-ML baseline track each other until 2016 and separate after it. The ML minus non-ML title gap is negative through 2016, turns positive in 2018, and reaches 1.81 per 100 words by 2025, so the divergence is recent and widening.

PubMed abstracts remain denser than NeurIPS abstracts throughout, 4.41 against 2.97 per 100 words in 2025, so machine learning is not yet at biomedical density in abstracts, though it passed PubMed in titles in 2023 and reached 4.15 against 2.59 by 2025. The comparison has a limit. Biomedical abstracts address a defined clinical readership, whereas machine learning is now read across statistics, engineering, the social sciences and policy, so an unfamiliar acronym costs more here than the density comparison alone suggests.

Reuse. Density is one cost and reuse is a second. Roughly 88 to 90 percent of ML acronyms appear fewer than ten times, against 79 percent science-wide, so ML acronyms are both more numerous and more likely to be novel to the reader. Single-use rates are comparable at 29 to 32 percent, and the difference falls in the 2 to 9 use bucket (Table 2). High-density PubMed acronyms are largely standardized vocabulary recurring across many papers, whereas ML acronyms remain paper-specific. P1 and P6 in Section 6 therefore target reuse and expansion rather than absolute density, which is already below the biomedical level.

4 Citations

Top and bottom deciles. Citation count is the field’s primary proxy for research impact, so if it correlates with readability the decline compounds, because papers harder to read reach a smaller audience and accumulate fewer citations. Counts come from Semantic Scholar, covering 99.8 percent of NeurIPS papers through 2024 (Appendix A).

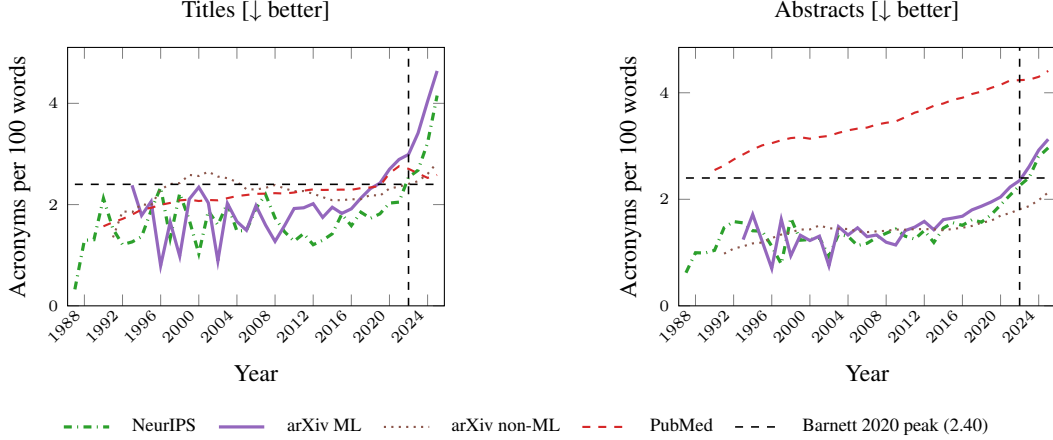


Figure 4: **Acronym density in titles (left) and abstracts (right), 1987 to 2025.** Lower values indicate fewer acronyms per word ($\downarrow$ better). Dashed black horizontal line marks the Barnett and Doubleday science-wide peak of 2.40 per 100 words (2019). NeurIPS and arXiv ML diverge from the non-ML control around 2017 on titles and around 2020 on abstracts.

Table 1: NeurIPS readability for top-10 percent vs bottom-10 percent cited papers (within-year citation percentile, pooled through 2024; the most recent year is excluded because its citation counts are immature). Values are mean with 95 percent confidence interval (normal approximation). Bold indicates the gap favours the top-cited group by at least 0.05 in magnitude.

Metric	Better	Top 10%	Bottom 10%	Δ (top – bottom)
Flesch Reading Ease	$\uparrow$ better	20.75 ± 0.47	18.66 ± 0.50	+2.09
Flesch-Kincaid Grade	$\downarrow$ better	15.83 ± 0.09	16.43 ± 0.09	-0.60
Gunning Fog Index	$\downarrow$ better	18.86 ± 0.11	19.55 ± 0.11	-0.69
SMOG Index	$\downarrow$ better	16.64 ± 0.07	17.11 ± 0.07	-0.48
Dale-Chall Score	$\downarrow$ better	13.09 ± 0.04	12.94 ± 0.04	+0.14
Spache Grade	$\downarrow$ better	7.82 ± 0.03	7.98 ± 0.03	-0.16
Coleman-Liau Index	$\downarrow$ better	16.69 ± 0.09	16.39 ± 0.10	+0.29
ARI	$\downarrow$ better	17.96 ± 0.11	18.28 ± 0.12	-0.32
Linsear Write	$\downarrow$ better	15.95 ± 0.15	16.42 ± 0.16	-0.47
LIX	$\downarrow$ better	61.68 ± 0.27	62.63 ± 0.28	-0.95
RIX	$\downarrow$ better	8.70 ± 0.08	9.10 ± 0.09	-0.40
FORCAST	$\downarrow$ better	12.76 ± 0.03	12.69 ± 0.03	+0.07
Powers-Sumner-Kearl Grade	$\downarrow$ better	8.32 ± 0.03	8.46 ± 0.03	-0.14
Words per sentence	$\downarrow$ better	21.99 ± 0.18	23.24 ± 0.20	-1.25
Syllables per word	$\downarrow$ better	1.94 ± 0.01	1.95 ± 0.01	-0.01
$n_{\text{top}} = 2,493, n_{\text{bottom}} = 2,436$ papers.				

Table 1 compares classical readability scores and two surface linguistic measures for the top 10 percent and bottom 10 percent cited NeurIPS papers, by within-year citation percentile pooled through 2024. On twelve of fifteen metrics the top-cited papers score in the easier-to-read direction. Hohmann and Connell [2025] find a comparable contrast in conservation biology, though on presentational features such as broad titles and motivational language rather than on readability formulas, and they note that the literature on readability and citations is itself divided.

Subfield composition. Machine learning entered new subfields over this window. If those subfields are both harder to read and more cited, the association above and the decline in Section 2 could be compositional rather than a change in how any given subject is written. We test this in three steps. We recover topics from the text itself, we add them to the citation regression as fixed effects, and we hold the topic mix constant and recompute the readability trend. Topics are derived from the cleaned titles and abstracts of the NeurIPS corpus alone, with no external taxonomy, by TF-IDF, truncated SVD and k-means following Weißer et al. [2020].

Subfield predicts both readability and citations. Table 3 in Appendix E lists the 15 topics ordered by readability, from Diffusion and generative models at 9.0 Flesch to Online learning and bandits at 23.0, a spread of 14.0 points. Compared with each other, the least readable topics are the most cited ($r = -0.53$). That is the reverse of the association within a topic, so pooling subfields can produce the opposite conclusion.

Controlling for topic weakens that association but does not remove it. The standardized Flesch coefficient falls from 0.049 to between 0.026 and 0.037, and its confidence interval excludes zero at every topic resolution we tried (Appendix E).

The composition of the field did change over this period. The share of NeurIPS papers in the four least readable topics rose from 7.2 percent in 2015 to 58.4 percent in 2025. That change does not explain the readability decline. Holding the topic mix at its 2015 composition still leaves a fall of 9.6 Flesch points, against 12.1 observed, so 21 percent of the recent decline is compositional and the rest happens inside topics.

Bibliography growth. Bibliographies have grown alongside the corpus. Mean reference list length per NeurIPS paper rose from roughly 11 entries in 1987 to a peak of 62.9 in 2022 and has held between 59 and 61 since (Figure 10). Papers for which Semantic Scholar holds no reference list are excluded, and coverage of recent years is lower than of earlier ones, 73 percent of NeurIPS 2025 against 97 percent for 2022, so the level after 2022 should be read with that caveat. The mean NeurIPS paper now cites roughly five times as many works as in 1987. A reference count records what a paper cites and does not distinguish the works its argument rests on from those that provide background, a distinction P3 in Section 6 asks authors to make explicit.

5 Model-Based Assessment of Writing Quality

Judge scores over time. An obvious objection is that AI-assisted writing and AI-assisted review will correct these trends without venue intervention. The measurements reported in this section do not support that expectation. We apply the LLM as judge readability protocol of Cachola et al. [2025], six open-weight instruction-tuned models under three prompt templates, to the 24,772 NeurIPS abstracts from 1987 to 2024, of which 24,754 returned a parseable score. The models are Gemma-3-27B-Instruct, Gemma-4-31B-Instruct, Llama-3.1-8B-Instruct, Mistral-7B-Instruct, Mixtral-8x7B-Instruct, and Qwen2.5-32B-Instruct. Each model’s score is standardized to its own 1987 to 2022 baseline, and full prompt text appears in Appendix D. We note that Cachola et al. [2025] reach a different conclusion from the same protocol. Scoring individual plain language summaries, they find that model ratings correspond to human readability ratings more closely than Flesch-Kincaid does, and recommend dropping that formula while keeping Dale-Chall and Coleman-Liau as the primary measures and adding models alongside them. We evaluate a different quantity, the direction of change across four decades, which accuracy on individual texts does not entail recovering.

Figure 5 shows the standardized scores under each prompt. All six models under all three prompts report readability as flat or gently improving, with an upward shift after 2022. This is the opposite direction to every one of the 15 classical metrics over the same corpus. Figure 12 in Appendix F shows the same data without averaging, one line per model.

Reliability of the panel. A venue adopting a model screener would ask whether it is stable and whether its judgments agree. The panel is highly repeatable under resampling, and the six models correlate with one another. The reliability of their mean exceeds the level conventionally read as acceptable, and the direction of the yearly trend is unchanged under every standardization baseline we tried. Appendix F reports all four properties with the supporting tables. This panel therefore passes every reliability check a program chair is likely to apply, and it still points the opposite way to the human-grounded reference. Agreement among judges measures how much they differ from one another, and cannot measure what they get wrong together.

Correlates of the judge score. To see what the judge correlates with, we compute the per-paper Spearman correlation between its readability score and every writing metric, over the 24,754 NeurIPS abstracts it scored (Figure 13, in Appendix F). The judge registers some components of reader effort and not others. It correlates negatively with sentence length and syntactic complexity, scoring those abstracts as less readable, in the same direction as the classical formulas, while correlating positively

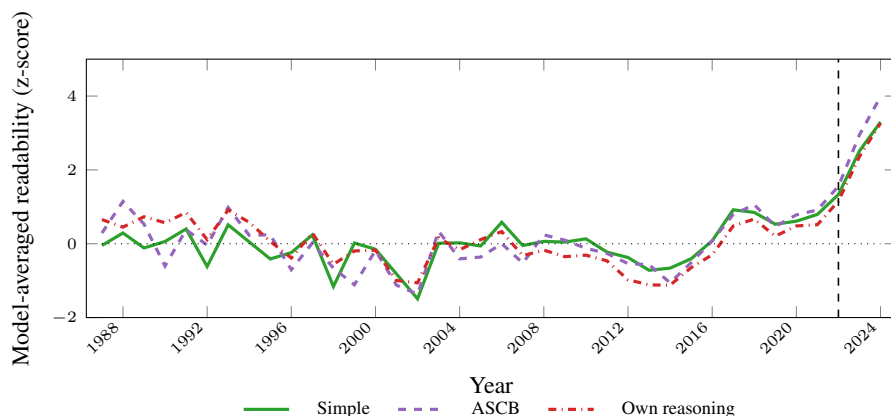


Figure 5: **LLM as judge readability on NeurIPS abstracts, 1987–2024.** Each curve is the mean z-score across all six open-weight models for one prompt variant, standardized to each model’s own 1987–2022 baseline (dotted line at zero). All three prompts agree: LLM-judged readability is approximately flat before 2022 and rises sharply after the availability of instruction-tuned writing assistants (vertical dashed line), directly contradicting the classical readability decline in Figure 2.

with noun phrase density, verb and noun density, and sensational language, and remaining near neutral on acronym density. Every correlation is small, under 0.25 in absolute value, and the pattern is consistent across all six models. These are correlations, not causal claims. What they show is that the judge co-varies with a different and partly opposed set of features than the human-grounded reference. P7 exists for this reason, since the measurement a venue enforces has to be one whose features can be inspected.

The implication for an AI-native venue is direct, and it applies at the two nodes where such a model would be deployed. At the reviewer node, a model that summarizes or drafts an assessment of writing quality does not track the same construct as the human-grounded metrics. At the program committee node, the same model used for desk-reject triage or score aggregation would rank submissions on features partly opposed to reader effort. In both cases the failure is silent, because the panel’s own agreement statistics report it as reliable. Supporting reviewers without replacing human accountability therefore requires an external reference against which the assistant can be checked.

6 Proposed Standards for AI-Native Venues

We propose seven standards, ordered by implementation cost. Each includes a measurable success criterion evaluable one year after a pilot. They are written for a venue that is integrating AI into its pipeline, and the organizing principle is that every standard is a transparent, auditable measurement rather than a black-box model judgment.

Each acts at a named point in that pipeline. P1, P2 and P5 are author-side checks run before submission. P3 acts at the citation node, asking which references a claim depends on. P6 and P7 are platform standards, a machine-readable artefact and the tooling that reads it. Only P7 requires the venue to build anything.

P1. Acronym budget and approved-term list. No abstract may introduce more than two novel acronyms. An acronym on a venue-maintained approved-term list does not count against the budget, and every novel acronym must be expanded on first use. The budget is the discipline medical journals apply through editorial review, and the approved-term list is what makes it enforceable without penalising the vocabulary a field already shares. Success criterion: median novel acronym count per abstract falls by 30 percent one year after the pilot.

P2. Readability threshold on human-grounded metrics. A submission-time report flags abstracts that fall below a threshold on Flesch Reading Ease, mean sentence length, and mean parse depth, which are transparent and grounded in human cognitive load. The threshold is soft in the first year

and anchored to a recent venue median, preventing further degradation rather than imposing an unprecedented standard. Authors who receive a flag may justify specialist vocabulary that cannot be simplified without loss of precision. Success criterion: mean abstract Flesch Reading Ease rises by five points.

P3. Stricter citation standards. Every paper identifies its three core citations, those without which the contribution cannot be evaluated, and every remaining citation must support a specific claim. Authors attest that every entry resolves to a retrievable paper. This targets both decorative citation and, in an AI-native pipeline, hallucinated references. Success criterion: mean citations per paper stabilizes or decreases.

P4. Standalone visual elements. Every accepted paper carries at least one figure that conveys the core contribution and is understandable without reading the full abstract, which lowers the barrier for interdisciplinary readers, the same audience widening noted in Section 3. Success criterion: the proportion of accepted papers with a standalone explanatory figure exceeds 80 percent.

P5. Plain language summary. Require a 100-word plain language summary written for a non-specialist reader, using no acronyms introduced only within the paper. The NEJM and the NIH public access policy both require such summaries, and this standard is no longer hypothetical in machine learning. ICML 2026 asks the authors of every accepted paper for a lay summary of at most 10 sentences or 200 words, with explicit guidance to avoid jargon [International Conference on Machine Learning, 2026]. P5 differs only in binding the summary to the acronym rule of P1 and in setting a measurable compliance target. Success criterion: 95 percent of accepted papers carry a compliant summary.

P6. Pre-registered acronym glossary. Every paper submits a machine-readable glossary of each acronym in the title, abstract, and body, with its expansion. The glossary makes P1 enforceable at scale, gives search and indexing an explicit expansion for every abbreviation, and seeds the community approved-term list. Success criterion: 100 percent of accepted papers ship a compliant glossary.

P7. Open source audit tooling. Every standard above is a measurement, so each should be computable by anyone the standard applies to. The tooling that computes them should be open source, independently reproducible, and integrated into the submission workflow, so that an author can check a draft before submitting it, a reviewer can verify a flag rather than take it on trust, and a program chair can monitor venue-level trends across a cycle. Commissioning and maintaining that tooling is what makes the preceding six standards enforceable, and it is the one proposal here that asks the venue to build something. Success criterion: a venue-adopted audit tool is integrated into the submission workflow and venue-level dashboards are published.

7 Alternative Views

“Acronyms are efficient for domain experts.” They are, within a community of practice, when they are reused. Table 2 shows that 89 percent of NeurIPS acronyms are used fewer than ten times, ten points above the science-wide baseline, so most never become shared vocabulary. P1 asks for a budget and a reuse registry, which is precisely the efficiency the objection seeks to preserve.

“This is just field growth.” The arXiv non-ML categories grow from the same infrastructure over the same window, yet sit well below machine learning on every readability metric and at lower acronym density (Figures 2 and 4). Growth in subfields is a separate confound, and holding the NeurIPS topic mix fixed still leaves 79 percent of the recent decline in place (Section 4).

“AI-assisted writing will correct these trends.” Section 5 is direct evidence against this. Instruction-tuned models do not detect the degradation the classical metrics reveal, and they report it in the opposite direction. Either the model is not a valid instrument for reading ease, or it favours the increasingly model assisted text of recent years [Liang et al., 2025b, Stureborg et al., 2024, Panickssery et al., 2024]. This design cannot separate the two, since the corpus does not record

273 provenance, and under both a model’s own judgment fails to track the human-grounded reference. It
274 is therefore not the instrument that will correct the trend.

275 **“Stricter citation rules will penalise comprehensive scholarship.”** P3 caps nothing. It changes
276 what the bibliography declares. Three identified core citations, followed by an unrestricted tail
277 of supporting ones, tell a reviewer which works the contribution depends on and which provide
278 background. The requirement is additive, and comprehensive scholarship already satisfies it.

279 **“Writing standards are gatekeeping that will exclude newcomers.”** This is the objection we take
280 most seriously, since any new requirement raises the cost of entry to a literature whose openness has
281 been a strength. The three standards carrying the most weight cut the other way: P1, P2 and P5 target
282 acronym density, readability and a plain language summary, each of which lowers the barrier for a
283 reader or author new to a subfield or not writing in a first language. They are author-side checks made
284 before submission, not grounds for rejecting unfamiliar work, and every success criterion is defined
285 over properties of the text rather than the identity or institution of the authors. We can show only that
286 they are designed to lower the barrier, not that they do. A pilot would have to measure submission
287 rates by group, and a standard that reduced them for any group should be revised or withdrawn.

288 **“Enforced standards will stifle creativity.”** The BMJ, the NEJM, and JMLR have enforced explicit
289 style guides for decades without measurable loss of output quality, because enforcement targets the
290 form of communication rather than its content. A related worry is that a threshold invites gaming. P2
291 is therefore soft in its first year and lets authors justify vocabulary that cannot be simplified without
292 loss of precision, and P7 keeps the measurement open to inspection.

293 8 Discussion

294 The empirical picture is consistent across metric families. NeurIPS abstracts have become harder
295 to read on every classical metric, acronym density has passed the science-wide figure while reuse
296 has lagged it, and more readable papers are more cited even within a subfield. Each of these trends
297 raises the cost of reading, and the volume of material to be read has grown with them. The tooling an
298 AI-native venue would naturally reach for to manage that volume does not register any of it, and the
299 gap between the machine judge and the human-grounded metrics widens after 2022. Taken together
300 these are the preconditions for the knowledge in the dark that Jeschke et al. [2019] describe.

301 That loop is the reason the screening step matters more than it appears to. If models help write the
302 text, models screen it, and the accepted output returns to the corpora on which the next generation is
303 trained, then every quality signal in the cycle is a model signal, and training on recursively generated
304 data has documented consequences for model behaviour [Shumailov et al., 2024]. We do not measure
305 that cycle here, and our result speaks only to the screening step. What it says about that step is that
306 the signal does not track the human reading cost the classical formulas estimate.

307 Several limitations bound the findings. The analysis covers titles and abstracts rather than full texts,
308 so the magnitude of the trend may differ for full documents, though the direction is unlikely to
309 reverse [Plavén-Sigra et al., 2017]. Series are indexed by publication date, so year-level effects
310 are approximately rather than sharply dated, which matters most around the late 2022 marker. The
311 corpora concentrate on English-language and Western-institution venues, and citation analyses are
312 restricted to NeurIPS. The judge experiment uses six open-weight models of roughly 7 to 32 billion
313 active parameters, and larger closed-weight models may recover more of the classical signal. The
314 standards have not been evaluated prospectively, and the strongest test of the central claim, human
315 readability ratings on a time-stratified sample, remains to be run.

316 Machine learning research has outgrown its communication norms, and it is automating its pipeline
317 before agreeing on what to measure. This evidence supports one narrow answer about how the
318 institutions of science should change once AI is embedded in authorship and review. Review at the
319 scale of tens of thousands of submissions will need automated screening. Each automated judgment
320 should be anchored to a measurement whose features can be inspected, and that anchor should be a
321 property of the text rather than the output of another model. Our panel illustrates the risk, passing
322 every reliability check while reporting a four-decade decline as an improvement. The seven standards
323 proposed here are anchors of that kind.

References

- Adrian Barnett and Zoe Doubleday. The growth of acronyms in the scientific literature. *eLife*, 9:e60080, 2020. doi: 10.7554/eLife.60080. URL <https://doi.org/10.7554/eLife.60080>.
- Leonard Berlin. TAC: AOITROMJA? (the acronym conundrum: advancing or impeding the readability of medical journal articles?). *Radiology*, 266(2):383–387, 2013. doi: 10.1148/radiol.12121776. URL <https://doi.org/10.1148/radiol.12121776>.
- Joydeep Biswas, Sheila Schoepp, Gautham Vasan, Anthony Oipari, Arthur Zhang, Zichao Hu, Sebastian Joseph, Matthew Lease, Junyi Jessy Li, Peter Stone, Kiri L. Wagstaff, Matthew E. Taylor, and Odest Chadwicke Jenkins. AI-assisted peer review at scale: The AAAI-26 AI review pilot, 2026. URL <https://arxiv.org/abs/2604.13940>.
- Bruce K. Britton, Shawn M. Glynn, Bonnie J. F. Meyer, and M. J. Penland. Effects of text structure on use of cognitive capacity during reading. *Journal of Educational Psychology*, 74(1):51–61, 1982. doi: 10.1037/0022-0663.74.1.51. URL <https://doi.org/10.1037/0022-0663.74.1.51>.
- Isabel Cachola, Daniel Khashabi, and Mark Dredze. Evaluating the evaluators: Are readability metrics good measures of readability? In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 24011–24027, Suzhou, China, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.emnlp-main.1225. URL <https://aclanthology.org/2025.emnlp-main.1225/>.
- Nuo Chen, Moming Duan, Andre Huikai Lin, Qian Wang, Jiaying Wu, and Bingsheng He. Position: The current AI conference model is unsustainable! diagnosing the crisis of centralized AI conference, 2025. URL <https://arxiv.org/abs/2508.04586>.
- Gregory Druck, Jose Luis Paredes, Bevin Benson, and Ethan Smith. More articles are now created by AI than humans. *Graphite industry report*, 2025. URL <https://graphite.io/five-percent-more-articles-are-now-created-by-ai-than-humans>.
- William H. DuBay. *The Principles of Readability*. Impact Information, Costa Mesa, CA, 2004.
- Rudolf Flesch. *How to Write Plain English: A Book for Lawyers and Consumers*. Harper and Row, New York, 1979. ISBN 0060112786.
- Rudolph Flesch. A new readability yardstick. *Journal of Applied Psychology*, 32(3):221–233, 1948. doi: 10.1037/h0057532.
- Jiawei Gu, Xuhui Jiang, Zhichao Shi, Hexiang Tan, Xuehao Zhai, Chengjin Xu, Wei Li, Yinghan Shen, Shengjie Ma, Honghao Liu, Saizhuo Wang, Kun Zhang, Zhouchi Lin, Bowen Zhang, Lionel Ni, Wen Gao, Yuanzhuo Wang, and Jian Guo. A survey on LLM-as-a-judge. *The Innovation*, 7(6):101253, 2026. doi: 10.1016/j.xinn.2025.101253.
- Mollie Hawkes Hohmann and Sean D. Connell. Comparing the writing styles of highly and rarely cited papers in conservation biology. *Biological Conservation*, 307:111125, 2025. doi: 10.1016/j.biocon.2025.111125. URL <https://doi.org/10.1016/j.biocon.2025.111125>.
- Mollie Hawkes Hohmann, Adrian G. Barnett, Neil King, and Sean D. Connell. The evolution of scientific writing: an analysis of 20 million abstracts over 70 years in health and medical science. *Scientometrics*, 130:3349–3366, 2025. doi: 10.1007/s11192-025-05353-8. URL <https://doi.org/10.1007/s11192-025-05353-8>.
- International Conference on Machine Learning. ICML 2026 lay summaries. ICML 2026 author guidance, 2026. URL <https://blog.icml.cc/2026/05/07/icml-2026-lay-summaries/>.
- Jonathan M. Jeschke, Sophie Lokatis, Isabelle Bartram, and Klement Tockner. Knowledge in the dark: scientific challenges and ways forward. *FACETS*, 4(1):423–441, 2019. doi: 10.1139/facets-2019-0007. URL <https://doi.org/10.1139/facets-2019-0007>.
- Weixin Liang, Yaohui Zhang, Mihai Codreanu, Jiayu Wang, Hancheng Cao, and James Zou. The widespread adoption of large language model-assisted writing across society. *Patterns*, 6(12):101366, 2025a. doi: 10.1016/j.patter.2025.101366.
- Weixin Liang, Yaohui Zhang, Zhengxuan Wu, Haley Lepp, Wenlong Ji, Xuandong Zhao, Hancheng Cao, Sheng Liu, Siyu He, Zhi Huang, Diyi Yang, Christopher Potts, Christopher D. Manning, and James Zou. Quantifying large language model usage in scientific papers. *Nature Human Behaviour*, 9(12):2599–2609, 2025b. doi: 10.1038/s41562-025-02273-8.

374 Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-Eval: NLG evaluation
375 using GPT-4 with better human alignment. In *Proceedings of the 2023 Conference on Empirical Methods*
376 *in Natural Language Processing*, pages 2511–2522, 2023. URL [https://aclanthology.org/2023.](https://aclanthology.org/2023.emnlp-main.153/)
377 [emnlp-main.153/](https://aclanthology.org/2023.emnlp-main.153/).

378 Neil Millar, Bojan Batalo, and Brian Budgell. Trends in the use of promotional language (hype) in abstracts of
379 successful national institutes of health grant applications, 1985–2020. *JAMA Network Open*, 5(8):e2228676,
380 2022. doi: 10.1001/jamanetworkopen.2022.28676. URL [https://doi.org/10.1001/jamanetworkopen.](https://doi.org/10.1001/jamanetworkopen.2022.28676)
381 [2022.28676](https://doi.org/10.1001/jamanetworkopen.2022.28676).

382 Arjun Panickssery, Samuel R. Bowman, and Shi Feng. LLM evaluators recognize and
383 favor their own generations. In *Advances in Neural Information Processing Systems*
384 37, 2024. URL [https://proceedings.neurips.cc/paper_files/paper/2024/hash/](https://proceedings.neurips.cc/paper_files/paper/2024/hash/7f1f0218e45f5414c79c0679633e47bc-Abstract-Conference.html)
385 [7f1f0218e45f5414c79c0679633e47bc-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2024/hash/7f1f0218e45f5414c79c0679633e47bc-Abstract-Conference.html).

386 Steven Pinker. *The Sense of Style: The Thinking Person’s Guide to Writing in the 21st Century*. Viking, 2014.

387 Pontus Plavén-Sigraý, Granville James Matheson, Björn Christian Schiffler, and William Hedley Thompson.
388 The readability of scientific texts is decreasing over time. *eLife*, 6:e27725, 2017. doi: 10.7554/eLife.27725.
389 URL <https://doi.org/10.7554/eLife.27725>.

390 Hillary C. Shulman, Graham N. Dixon, Odalys M. Bullock, and Dolores Colón Amill. The effects of jargon
391 on processing fluency, self-perceptions, and scientific engagement. *Journal of Language and Social Psy-*
392 *chology*, 39(5–6):579–597, 2020. doi: 10.1177/0261927X20902177. URL [https://doi.org/10.1177/](https://doi.org/10.1177/0261927X20902177)
393 [0261927X20902177](https://doi.org/10.1177/0261927X20902177).

394 Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. AI models
395 collapse when trained on recursively generated data. *Nature*, 631(8022):755–759, 2024. doi: 10.1038/
396 [s41586-024-07566-y](https://doi.org/10.1038/s41586-024-07566-y).

397 Rickard Stureborg, Dimitris Alikaniotis, and Yoshi Suhara. Large language models are inconsistent and biased
398 evaluators, 2024. URL <https://arxiv.org/abs/2405.01724>.

399 John Sweller. Cognitive load theory, learning difficulty, and instructional design. *Learning and Instruction*, 4(4):
400 295–312, 1994. doi: 10.1016/0959-4752(94)90003-5. URL [https://doi.org/10.1016/0959-4752\(94\)](https://doi.org/10.1016/0959-4752(94)90003-5)
401 [90003-5](https://doi.org/10.1016/0959-4752(94)90003-5).

402 Tim Weißer, Till Saßmannshausen, Dennis Ohrndorf, Peter Burggräf, and Johannes Wagner. A clustering
403 approach for topic filtering within systematic literature reviews. *MethodsX*, 7:100831, 2020. doi: 10.1016/j.
404 [mex.2020.100831](https://doi.org/10.1016/j.mex.2020.100831).

405 Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan
406 Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with
407 MT-bench and Chatbot Arena. In *Advances in Neural Information Processing Systems 36: Datasets and*
408 *Benchmarks Track*, 2023. URL <https://arxiv.org/abs/2306.05685>.

409 Appendix

410 A Dataset

411 arXiv is processed from the Cornell-University/arxiv Kaggle snapshot
412 (arxiv-metadata-oai-snapshot.json) and split into per-year files. NeurIPS is scraped
413 from papers.nips.cc, with pre-2008 abstracts patched from proceedings.neurips.cc because
414 the early conference pages bundle full text into a single raw field. PubMed records are downloaded
415 from the NLM baseline FTP (<https://ftp.ncbi.nlm.nih.gov/pubmed/baseline/>) as
416 gzipped XML and parsed for English-language abstracts of at least six words. Every source
417 is normalized to one Pydantic-validated schema with fields paper_id, venue, year, title,
418 abstract, authors, arxiv_primary_category, and arxiv_categories. Per-source counts
419 at the 2026-04 snapshot are 2,823,623 arXiv papers (1991 to 2025), 30,595 NeurIPS papers
420 (1987 to 2025), and 24,463,208 PubMed papers (1990 to 2025). The arXiv 2025 partition covers
421 approximately three quarters of the calendar year. Citation counts are matched to Semantic Scholar
422 by normalized title, supplemented for 2.2 percent of papers by fuzzy-title and OpenAlex matches,
423 giving coverage of 99.8 percent of NeurIPS papers through 2024.

424 B Measurement Details

425 The metrics are chosen because each stands in for a cost a reader pays. Britton et al. [1982] show that
426 simplifying syntax reduces the cognitive capacity a reader expends, the mechanism cognitive load
427 theory describes [Sweller, 1994]. Dense noun phrases force readers to hold more concepts in parallel
428 [Pinker, 2014], and unfamiliar jargon disrupts fluent processing even where it is defined [Shulman
429 et al., 2020].

430 Classical readability is computed with the textstat library, which supplies the sentence splitting
431 and syllable counting the formulas take as inputs. The style metrics and the linguistic complexity
432 measures use spaCy with the en_core_web_sm model. Acronyms are extracted by a faithful port of
433 the algorithm of Barnett and Doubleday [2020], including its preprocessing for roman numerals, units
434 and chromosome names, so ML text is scored by exactly the rule that produced the science-wide
435 baseline we compare against. We have not benchmarked that rule against hand-labelled ML abstracts,
436 and dataset and model names are treated as acronyms wherever they satisfy it.

437 Sensational language uses the eight hype categories of Millar et al. [2022] together with a ninth of
438 further terms added by Hohmann et al. [2025], and we have not adapted the word lists beyond that.
439 The Millar lexicon was derived from NIH grant abstracts, so terms that are descriptive in machine
440 learning may be counted as promotional. Since the same lexicon is applied to every venue and year,
441 this affects the level of the measure rather than the direction of its change, which is what Section 2
442 reports.

443 Abstract length grows over the period, but the classical formulas take ratios as inputs, words per
444 sentence and syllables per word, so a longer abstract does not mechanically lower its score. Length
445 enters the citation model above as an explicit control.

446 C Additional Figures and Tables

447 Table 2 gives the full acronym reuse distribution by venue, referenced from Section 3.

448 Figure 6 and Figure 7 show all 15 classical readability metrics, Figure 8 the nine writing style metrics,
449 and Figure 9 the ten sensational language categories separately, each on NeurIPS, arXiv ML, arXiv
450 non-ML and PubMed. Figure 10 reports mean bibliography length per NeurIPS paper and Figure 11
451 annual paper counts, both referenced from the body.

452 D LLM as Judge Prompts and Models

453 **Prompts.** We use the three prompt templates from Cachola et al. [2025] (their Table 10). The in-
454 struction text is reproduced verbatim. The only deviation is that we omit the Reason: <reasoning>
455 output field, retaining only the numeric score, to reduce token consumption. All three share a 1 to

Table 2: Acronym frequency distribution by venue (abstract acronyms). Each cell shows the percentage of unique acronyms appearing in that frequency range. B&D baseline: Barnett and Doubleday (2020), all scientific fields 1950–2019. Single-use rates are comparable across venues, but ML venues carry a larger 2–9 use bucket, so 88 to 90 percent of ML acronyms appear fewer than ten times against 79 percent science-wide.

Venue	Unique acronyms	1	2–9	10–99	100–999	1000+
NeurIPS	15,693	31.2%	58.3%	9.5%	1.0%	0.0%
arXiv ML	115,229	28.9%	59.1%	10.3%	1.5%	0.2%
arXiv non-ML	181,570	32.0%	50.7%	13.7%	3.1%	0.6%
PubMed	1,692,497	31.1%	48.5%	16.4%	3.1%	0.9%
B&D (2020)	1,112,345	30.0%	49.0%	—	—	—

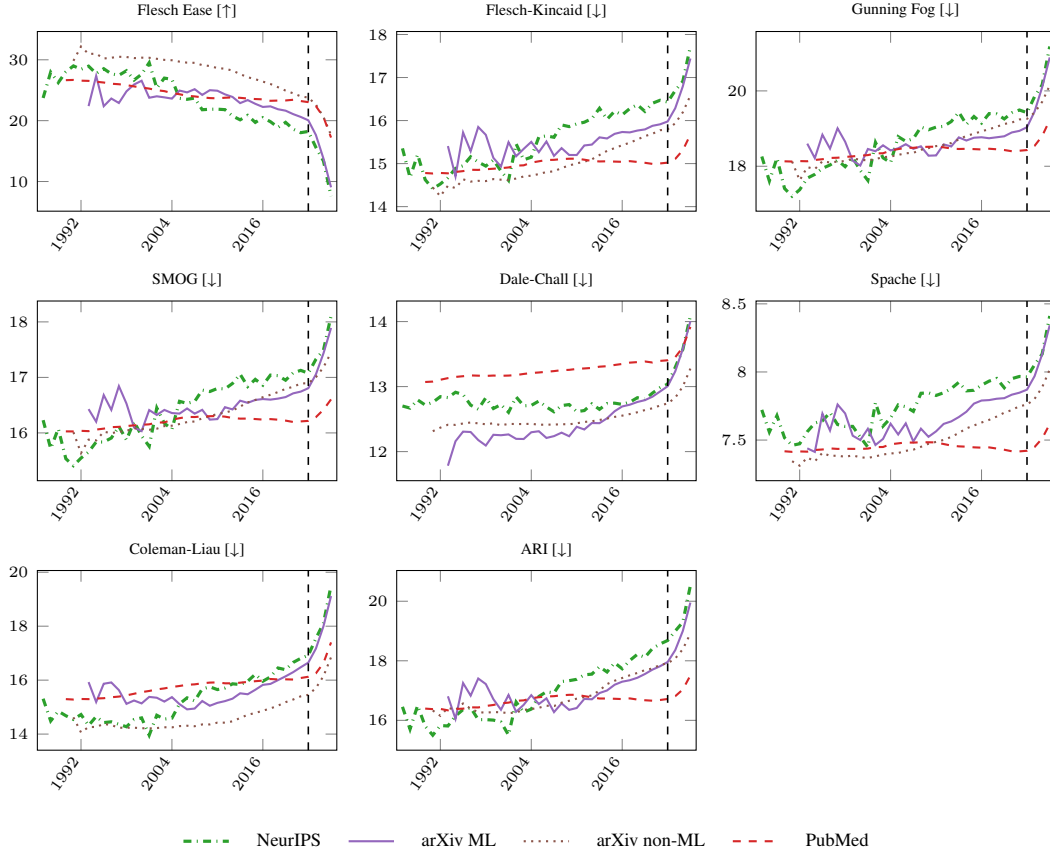


Figure 6: **Classical readability metrics, part A: Flesch family and grade-level formulas, 1987–2025.** Arrows indicate the direction of easier reading ($\uparrow$ = higher is better; $\downarrow$ = lower is better). Every grade-level metric rises on NeurIPS and arXiv ML after 2015 and accelerates after 2020; Flesch Reading Ease drops correspondingly. Vertical dashed lines mark the late-2022 availability of instruction-tuned writing assistants. Part B with the remaining seven metrics appears in Figure 7.

456 5 scale where 1 requires expert background knowledge and 5 is readable to the general population.
457 The simple prompt gives the bare scale, the ascb prompt adds American Society for Cell Biology
458 science-communication guidelines, and the own_reasoning prompt instructs the model to use its
459 own judgment and not to use readability formulas.

460 **Models and decoding.** We run six open-weight instruction-tuned models: Gemma-3-27B-Instruct,
461 Gemma-4-31B-Instruct, Llama-3.1-8B-Instruct, Mistral-7B-Instruct (v0.3), Mixtral-8x7B-Instruct,
462 and Qwen2.5-32B-Instruct. Decoding uses temperature 0.7 with sampling enabled and a cap of 8

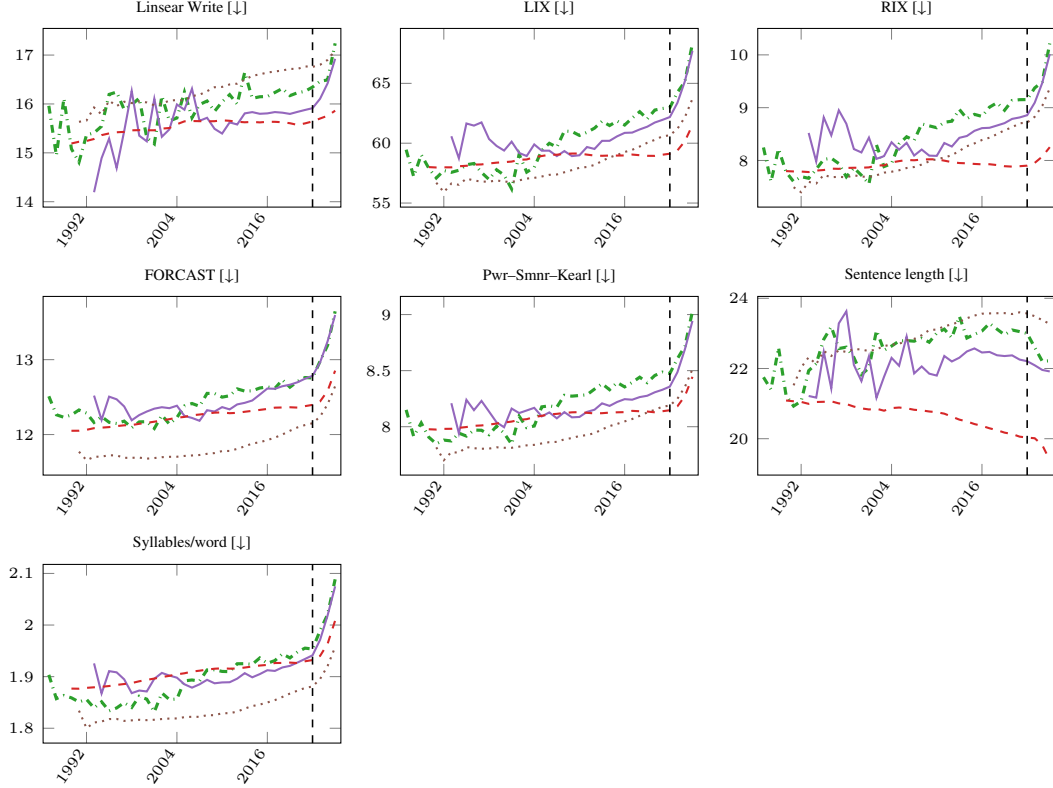


Figure 7: **Classical readability metrics, part B: Linsear Write, LIX, RIX, FORCAST, Powers-Sumner-Kearl, sentence length, and syllables per word, 1987–2025.** Arrows indicate the direction of easier reading ($\downarrow$ = lower is better). Vertical dashed lines mark the late-2022 availability of instruction-tuned writing assistants. Part A with the Flesch family and the first eight grade-level formulas appears in Figure 6.

new tokens per generation. We run three sampled generations per (paper, prompt, model) triple. Each model’s yearly means are standardized to its own 1987 to 2022 mean and standard deviation, and the figure reports the average of the six z-scored series.

E Topic Controls

Deriving the topics. NeurIPS papers carry no subject labels, so topics are derived from the papers themselves. We clean the title and abstract of each of the 30,595 NeurIPS papers, resolving LaTeX and removing URLs, represent them as TF-IDF vectors over words and word pairs, reduce to 150 dimensions by truncated singular value decomposition, and cluster with k-means, following Weißer et al. [2020]. No external subject taxonomy enters at any point, and no other venue enters at any point. To check that the clusters are not an artefact of one sample, we cluster two disjoint thirds of the corpus separately and compare their labels on the held-out third. Agreement is well above chance, with adjusted Rand index 0.37 and adjusted mutual information 0.53 at $k = 15$. Table 3 lists the resulting topics with the terms they were read from.

Effect on the citation association. The citation model regresses $\log(1 + \text{citations})$ on Flesch Reading Ease, controlling for log abstract length and log author count, with year fixed effects absorbed by within-transformation and standard errors clustered by year (CR1). All continuous variables are z-scored before the transformation, so the reported coefficient is standardised. The sample is the 24,725 NeurIPS papers matched to Semantic Scholar through 2024 in 38 year clusters. Year fixed effects also absorb paper age, since citations are compared only within a publication year.

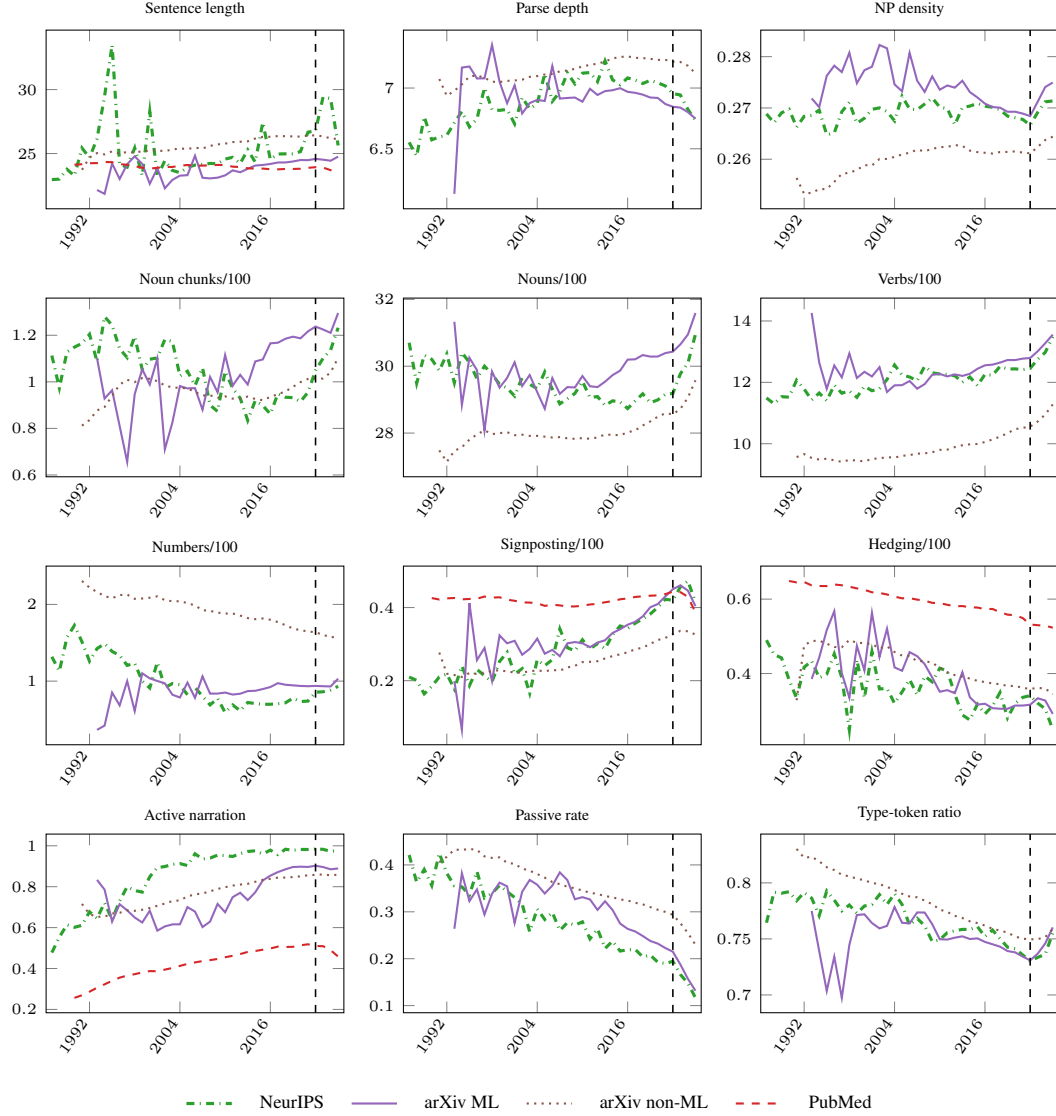


Figure 8: **Hohmann writing style metrics on NeurIPS, arXiv ML, arXiv non-ML, and PubMed, 1987–2025.** PubMed data are available only for the four metrics computed by lightweight regex and word-level rules (sentence length, signposting, hedging, and active narration). The remaining metrics depend on full spaCy dependency parsing and were not run on the 24.5 million paper PubMed corpus because of compute cost, so the corresponding panels show NeurIPS, arXiv ML, and arXiv non-ML only. Across these metrics, ML and non-ML venues are largely parallel. Only signposting, active narration, and passive rate show a clear widening of the ML versus non-ML gap after 2015.

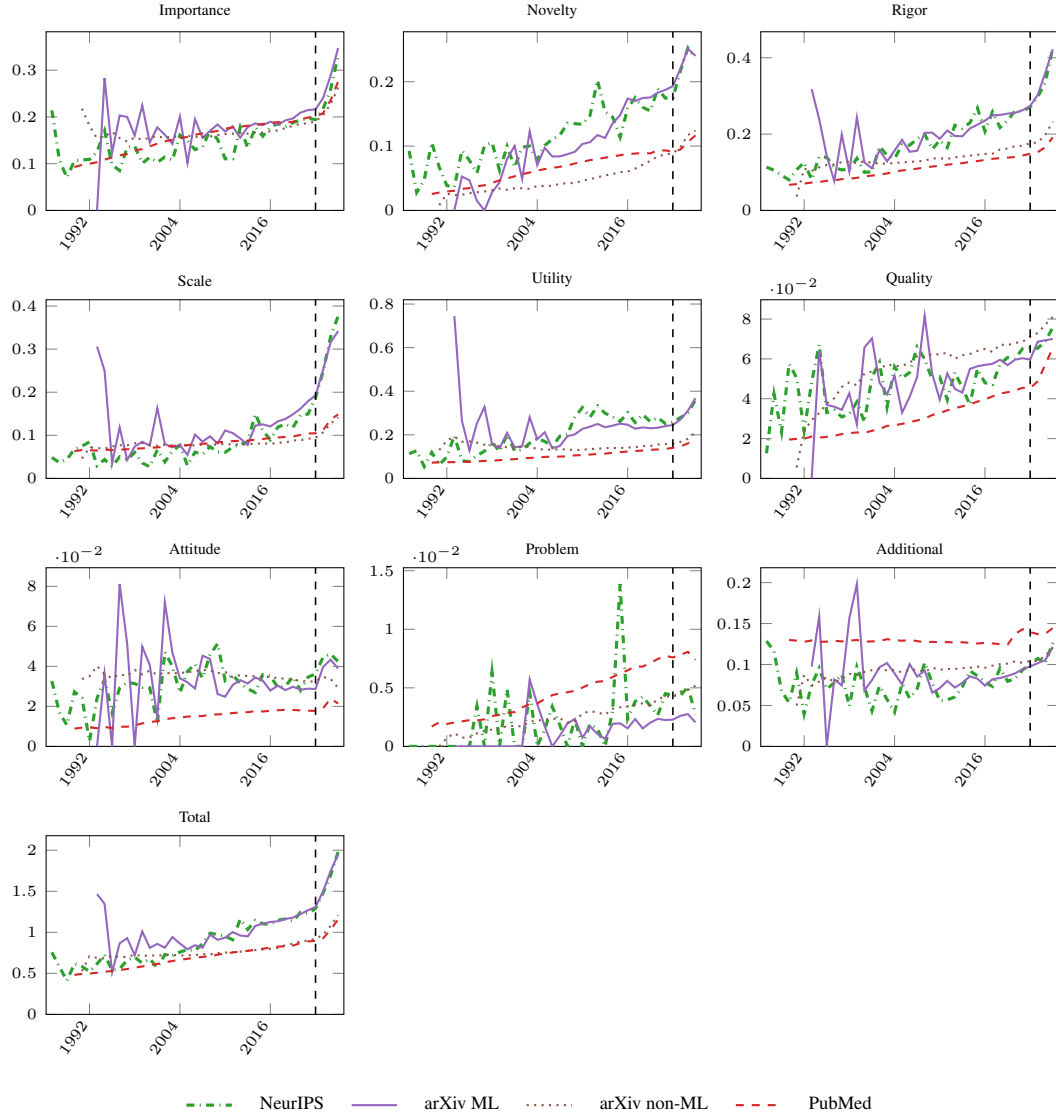


Figure 9: **All nine sensational language categories per 100 abstract words, 1987–2025.** Novelty and scale rise sharply after 2015, reaching roughly 2.1 times and 3.4 times their 2015 levels on NeurIPS by 2025 (1.6 and 2.7 times on arXiv ML); the total rate and rigor rise more modestly. Vertical dashed lines mark the late 2022 availability of instruction-tuned writing assistants.

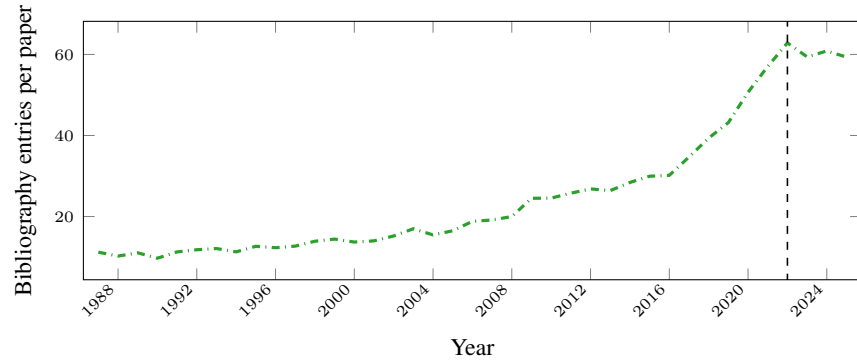


Figure 10: **Mean bibliography length per NeurIPS paper, 1987–2025.** Reference list length grew from roughly 11 entries per paper in 1987 to approximately 63 entries in 2022, a fivefold to sixfold increase over 35 years. Mean bibliography length then plateaued across 2023 to 2025 (59.4, 60.9, and 59.4 entries). Vertical dashed line marks the late 2022 availability of instruction-tuned writing assistants.

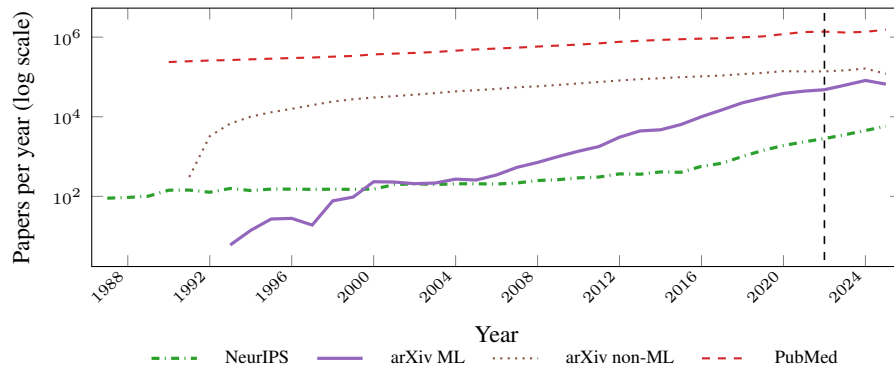


Figure 11: **Papers per year, 1987 to 2025.** arXiv ML submissions grew by roughly four orders of magnitude between 1993 and 2025. NeurIPS grew by a factor of roughly 65 between 1987 and 2025, from fewer than 100 accepted papers to approximately 5,800. PubMed and arXiv non-ML show sustained but slower growth over the same period. The volume increase is the structural backdrop against which the readability and acronym trends in Sections 2 and 3 must be interpreted.

482 We add the topics to that regression as fixed effects, at four resolutions (k of 10, 15, 20 and 30).
 483 Subfield explains a substantial share of citation variance. The within R^2 rises from 0.066 without
 484 topic effects to between 0.156 and 0.165 with them. The readability coefficient falls once subfield is
 485 held constant, from 0.049 to between 0.026 and 0.037, but its confidence interval stays clear of zero
 486 at every resolution, with a lower bound no smaller than 0.009. Readability is therefore associated
 487 with citations within subfields, not only between them.

488 **Holding the topic mix constant.** The counterfactual in Section 4 reweights each year to the topic
 489 mix of a base year and recomputes mean readability. This separates the decline caused by the field
 490 moving into less readable subfields from the decline occurring inside each subfield. We run it from
 491 2015 rather than from the start of the corpus for two reasons. Starting at the base year makes the
 492 observed and reweighted series coincide at the first point, so the two declines are compared directly.
 493 And the early years are too thin to reweight: NeurIPS 2002 holds 207 papers spread across 14 topics,
 494 nine of them with fewer than ten papers, so its reweighted mean is unstable. Starting in 2002 reverses
 495 the sign of the composition share on that instability alone, which is an artefact rather than a result.
 496 The choice of base year does not drive the conclusion. On a 2010 base the composition share is 12
 497 percent, against 21 percent on 2015, so most of the decline is within-topic on either.

Topic	Highest-weight terms	Papers	Flesch	Median cit.
Diffusion and generative models	diffusion, diffusion models	1,269	9.0	34
Vision and language	language, visual	1,810	11.2	35
Large language models	llms, language	2,141	13.3	40
Applied methods and benchmarks	data, training	4,224	13.9	25
3D vision and reconstruction	3d, image	1,540	14.0	33
Adversarial robustness	adversarial, attacks	808	14.1	35
Reinforcement learning	reinforcement, reinforcement learning	2,271	15.2	27
Optimization theory	gradient, convergence	1,565	15.2	28
Bayesian inference	inference, bayesian	2,134	16.4	27
Computational neuroscience	human, visual	1,813	18.1	16
Graph neural networks	graph, graphs	1,220	18.5	30
Object detection	object, objects	1,186	18.7	38
Neural network architectures	neural, networks	2,993	20.7	29
Learning theory and algorithms	data, algorithm	3,500	20.8	28
Online learning and bandits	regret, algorithm	2,121	23.0	17

Table 3: **NeurIPS topics, readability and citation impact.** The 15 topics recovered from the cleaned titles and abstracts of 30,595 NeurIPS papers, ordered from least to most readable. Topic names are our reading of each cluster and are printed beside the highest-weight terms they were read from. Flesch is the mean Flesch Reading Ease of the topic’s abstracts, and Median cit. the median citation count over the papers in the regression sample. Readability spans 14.0 points across topics, and the least readable topics are the most cited ($r = -0.53$ across topics), the opposite sign to the within-topic association reported in Appendix E.

Model	1987	2024	Δ	SD	Same
Gemma-3-27B	2.14	2.17	+0.03	0.02	98%
Gemma-4-31B	1.57	1.67	+0.10	0.03	95%
Llama-3.1-8B	1.97	2.20	+0.24	0.15	75%
Mistral-7B	2.62	3.04	+0.41	0.21	44%
Mixtral-8x7B	2.81	3.26	+0.44	0.09	83%
Qwen2.5-32B	2.18	2.10	-0.09	0.04	92%

Table 4: **Raw judge scores and run-to-run spread.** Judge score on the 1 to 5 scale under the simple prompt, in 1987 and 2024, with the change Δ . SD is the mean standard deviation across the three sampled generations for a given abstract, and Same is the proportion of abstracts scored identically on all three runs. The 1987 column rests on 59 to 87 scored abstracts depending on model, against 4,493 in 2024, so the per-model Δ is estimated far less precisely at the 1987 endpoint than at the 2024 one.

F Judge Panel Reliability

Table 4 reports the raw judge scores and the spread between sampled runs, and Table 5 the agreement between judges, both referenced from Section 5. The agreement statistics use the abstracts scored by all six models. The others use each model’s own scored abstracts.

Two further properties bound what the panel can register. Re-standardizing on 1987 to 2010, 1987 to 2016, or the full period places the 2024 model-averaged z between 2.48 and 4.47, against 3.17 to 3.87 for the 1987 to 2022 window used in the body, so the direction is unchanged under all four windows. The judges also occupy a narrow band of the response scale, with 97 to 99 percent of abstracts receiving a six-model mean that rounds to 2 or 3, so the panel resolves roughly two levels rather than five. An instrument with that resolution has little room to register a slow change in either direction, which is a further reason not to read a flat judge series as evidence that the underlying property held steady.

Prompt	mean ρ	min	max	α
simple	0.396	0.286	0.487	0.786
ascb	0.370	0.241	0.478	0.754
own_reasoning	0.409	0.296	0.496	0.791

Table 5: **Inter-judge agreement.** Per-paper Spearman ρ over the 15 model pairs, and Cronbach’s α over the six judges, on the abstracts scored by all six ($n = 23,257$ to $24,609$ depending on prompt).

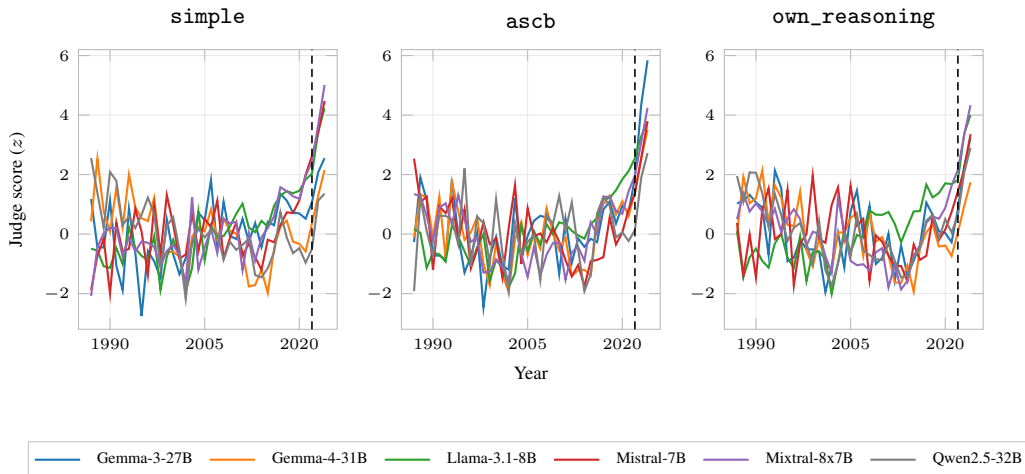


Figure 12: **Per-model judge scores by prompt template, standardized separately.** Each panel is one prompt template and each line one model, standardized to its own 1987 to 2022 mean and standard deviation. The dashed line marks late 2022. For each combination we fit a straight line through its yearly scores. The slope of that line is the average change in z per year. Before 2022 the slopes point in no consistent direction, nine of the 18 upward and nine downward, and none is steeper than $0.08 z$ per year, which is small against a year-to-year spread of about $4 z$. After 2022 all 18 point the same way, rising by between 1.24 and $3.94 z$. The model-averaged series in Figure 5 does not show the pre-2022 level differences or the threefold spread in the size of the rise, which follows from the differing baseline standard deviations.

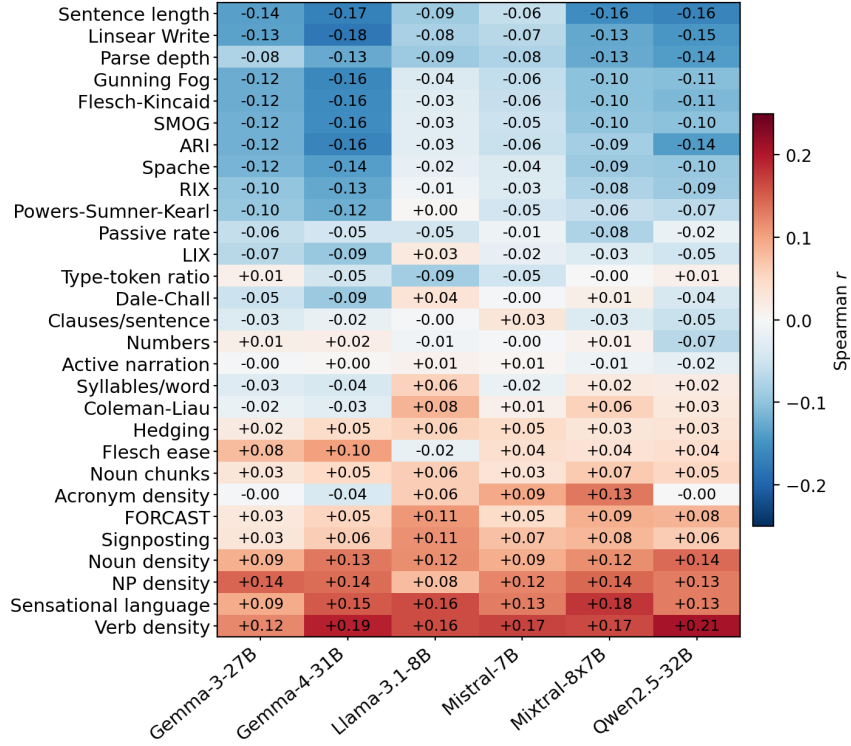


Figure 13: **What the LLM judge correlates with.** Per-paper Spearman correlation between each writing metric and each model’s judge score, over 24,754 NeurIPS abstracts. A model’s score for an abstract is the median of its nine generations, three prompts by three sampled runs. Rows are sorted by the correlation with the mean of those six per-model scores (1987–2024, the years the judge was scored). The judge score runs 1 to 5 with higher meaning easier, so each cell’s sign says which way that feature moves the judge. **Red (positive): more of that feature is judged *more* readable**, as for verb density, sensational language and noun phrase density. **Blue (negative): more of it is judged *less* readable**, as for sentence length and syntactic complexity, which is the direction the classical formulas also take. Near-white cells, including acronym density, indicate near indifference. These are correlations, not causal claims, and every effect is small ($|r| < 0.25$).